

Timing: 12:00 AM to 12:45 PM

Max mark: 10

Instruction: 1, 2 are mandatory; choose either 3 or 4. No step jump.

1. Thoroughly derive the Cauchy-Riemann (CR) equations. 4
2. Evaluate $\int_c \frac{e^{3iz}}{(z+\pi)^3} dz$ where c is the circle $|z - \pi| = 3.2$. 2
3. Evaluate $\int_0^{1+i} (x^2 - i y) dz$ along the path
 - (a) $y = x$
 - (b) $y = x^2$Also comment on the nature of the function $x^2 - iy$. 4
4. (a) Derive polar form of CR relation. 2
(b) check if $f(z) = \frac{1}{z^n}; z \neq 0$ is differentiable or not. 2